



Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Pre-Processing + SVM + Decision Tree

**MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA**

**Inteligência Artificial para as Telecomunicações
2025/2026**



Objectives

- The primary goal of this task is **to create a comprehensive and efficient workflow** tailored specifically for pre-processing the **Wine Quality** dataset.
- The final result will be used to train a **SVM** model, in order to **classify** wine's quality as **high** or **low**.

Dataset

wine_quality_learning

- A **.csv file**: This file contains information about the **various properties of the** wine, such as acidity, alcohol, pH, etc. Specific for **training** the model.

winequality_test

- A **.csv file**: This file contains information about the **various properties of the** wine, such as acidity, alcohol, pH, etc. Specific for **testing** the model.

Both share the exact same features.

Dataset - Features

12 attributes:

- **fixed acidity**: Total amount of organic acids found in wine.
- **volatile acidity**: Quantity of acetic acid generated by yeast or bacteria.
- **citric acid**: The quantity of citric acid.
- **residual sugar**: The sugar remaining in wine after fermentation.
- **chlorides**: Amount of salt (chlorides) in the wine.
- **free sulfur dioxide**: Amount of free sulfur dioxide in the wine.
- **total sulfur dioxide**: Total amount of sulfur dioxide in the wine.
- **density**: Wine's density, related to the alcohol and sugar content.
- **pH**: Measures the wine's acidity or alkalinity. A lower pH indicates higher acidity.
- **sulphates**: Sulphates quantity, which act as a preservative and antioxidant.
- **alcohol**: Percentage of alcohol in the wine.
- **quality**: The quality rating of the wine, ranging from 0 (poor) to 10 (excellent).

Dataset - Features

- **fixed acidity:** This is the total amount of organic acids found in wine, primarily succinic, citric, malic, and tartaric acids. The wine's stability, color, and flavor are all influenced by these acids. Depending on the grape variety and climate, wine's average fixed acidity can range from 4 to 12 grams per liter (g/L).
- **volatile acidity:** This is the quantity of acetic acid generated by yeast or bacteria during age or fermentation. Acetic acid adds to the degree of sourness and pungency. It is optimal for wine to have an average volatile acidity level of less than 0.4 g/L. When volatile acidity above the sensory threshold of 0.6 to 0.9 g/L for red wines and 0.5 to 0.7 g/L for white wines, it is deemed a wine defect.
- **citric acid:** Adds a fresh, citrus-like acidity to the wine. Although present in small amounts, it contributes to the wine's overall acidity and liveliness. The quantity of citric acid is usually very low and needs to be measured in decimal points.

Dataset - Features

- **residual sugar:** Refers to the sugar remaining in wine after fermentation, affecting its sweetness. Dry wines have low residual sugar, while sweeter wines have higher levels.
- **chlorides:** Measures the amount of salt (chlorides) in the wine. High levels can negatively affect the flavor.
- **free sulfur dioxide:** Refers to the amount of free sulfur dioxide in the wine, which protects it from oxidation and microbial growth.
- **total sulfur dioxide:** Represents the total amount of sulfur dioxide in the wine, including both free and bound forms. High levels can affect flavor and safety.

Dataset - Features

- **density:** Refers to the wine's density, which is related to the alcohol and sugar content. Density is measured as a continuous variable, often requiring decimals for accuracy. A double type handles this range of precise values effectively.
- **pH:** Measures the wine's acidity or alkalinity. A lower pH indicates higher acidity.
- **sulphates :** Refers to sulphates, which act as a preservative and antioxidant.

Dataset - Features

- **alcohol (double type):** Represents the percentage of alcohol in the wine, which influences its body and taste.
- **quality (string type):** The quality rating of the wine, usually based on sensory data, ranging from 0 (poor) to 10 (excellent).

Data preparation for **training** and **testing** must be **consistent**. This means that transformations applied to the training set must be applied **equally** to the test set.

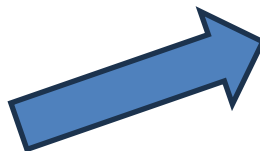
Pre-Processing

- Load both the training and the test dataset, **separately**.

Training



Testing



Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from Local File System

Mode ☒ File ☐ Files in folder

File C:\Users\Ana Sousa\Downloads\WineQuality\winequality_learning.csv Browse...

Reader options

Format

Autodetect format

Column delimiter , Row delimiter ☒ Line break ☐ Custom \n

Quote char " Quote escape char \"

Comment char #

☒ Has column header ☐ Has RowID

☐ Support short data rows ☐ Prepend file index to RowID

Preview

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	D fixed a...	D volatile ...	D citric acid	D residual...	D chlorides	D free su...	D total su...	D density	D pH	D sulphates	D alcohol	S qual
Row0	7.4	0.7	0	1.9	0.076	11	34	0.998	3.51	0.56	9.4	=5

OK Apply Cancel ?

Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from Local File System

Mode ☒ File ☐ Files in folder

File C:\Users\Ana Sousa\Downloads\WineQuality\winequality_test.csv Browse...

Reader options

Format

Autodetect format

Column delimiter , Row delimiter ☒ Line break ☐ Custom \n

Quote char " Quote escape char \"

Comment char #

☒ Has column header ☐ Has RowID

☐ Support short data rows ☐ Prepend file index to RowID

Preview

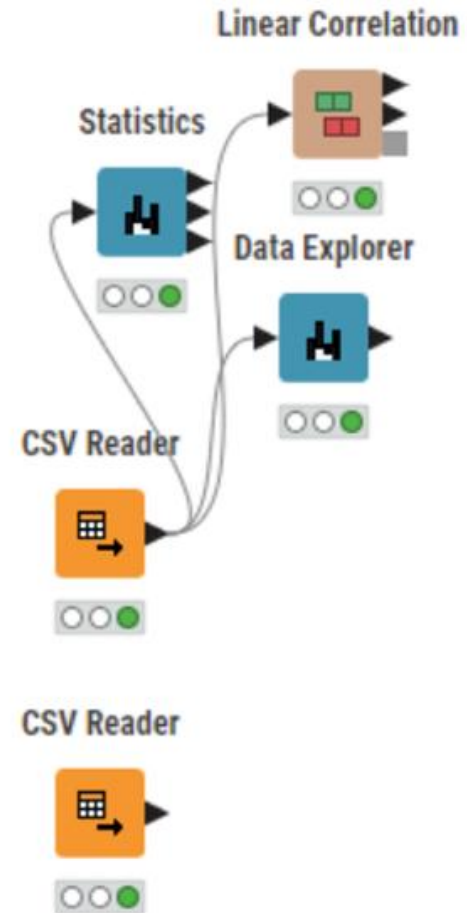
The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	D fixed a...	D volatile ...	D citric acid	D residual...	D chlorides	D free su...	D total su...	D density	D pH	D sulphates	D alcohol	S qual
Row0	6.9	0.74	0.03	2.3	0.054	7	16	0.995	3.45	0.63	11.5	=6

OK Apply Cancel ?

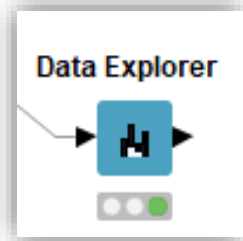
Pre-Processing

- Start by using data exploration nodes.



Data Preparation – Data Exploration nodes

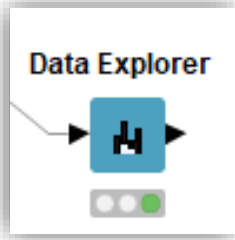
- Offers a range of options for displaying properties of the input data in an interactive view.



Column	Exclude Column	Minimum	Maximum	Mean	Standard Deviation	Variance	Skewness	Kurtosis	Overall Sum
+ fixed acidity	☐	4.600	15.900	8.416	1.742	3.035	0.940	1.084	12616.000
+ volatile acidity	☐	0.120	1.580	0.526	0.180	0.032	0.720	1.346	788.565
+ citric acid	☐	0	1	0.275	0.195	0.038	0.275	-0.811	412.440
+ residual sugar	☐	0.900	15.500	2.544	1.405	1.975	4.471	28.063	3813.550
+ chlorides	☐	0.012	0.611	0.088	0.048	0.002	5.622	40.506	132.308
+ free sulfur dioxide	☐	1	72	15.614	10.464	109.503	1.274	2.005	23406

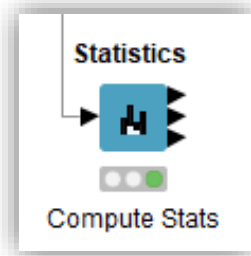
Data Preparation – Data Explorer

- Offers a range of options for displaying properties of the input data in an interactive view.

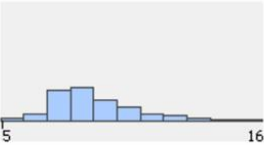
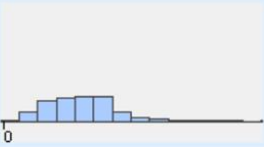
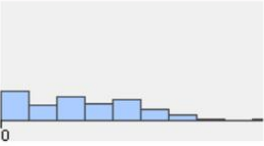


Numeric Nominal Data Preview					
Column	Exclude Column	No. missings	Unique values	All nominal values	Frequency Bar Chart
quality	☐	0	6	=5, =6, =7, =4, =8, =3	

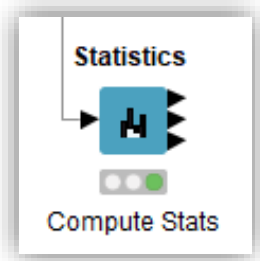
Data Preparation – Statistics



Offers a range of **statistical measures**, such as mean, median, standard deviation, minimum, maximum, quantiles, and more.

Column	Min	Mean	Median	Max	Std. Dev.	Skewness	Kurtosis	No. Missing	No. +∞	No. -∞	Histogram
fixed acidity	4,6	8,4163	?	15,9	1,7421	0,9399	1,0835	0	0	0	
volatile acidity	0,12	0,5261	?	1,58	0,1798	0,7197	1,3463	0	0	0	
citric acid	0.0	0,2751	?	1	0,1955	0,2753	-0,8112	0	0	0	
residual sugar	0,9	2,5441	?	15,5	1,4052	4,4708	28,0633	0	0	0	

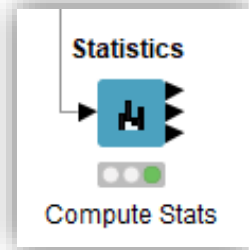
Data Preparation – Statistics



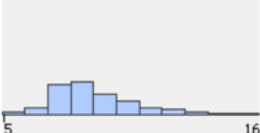
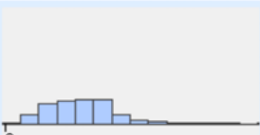
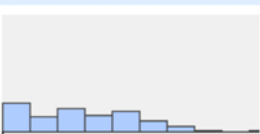
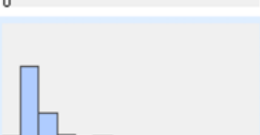
Search:

Column	Exclude Column	No. missings	Unique values	All nominal values	Frequency Bar Chart
quality	☐	0	6	=5, =6, =7, =4, =8, =3	

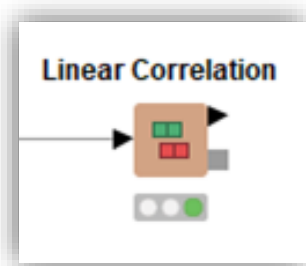
Data Preparation – Statistics



Offers a range of **statistical measures**, such as mean, median, standard deviation, minimum, maximum, quantiles, and more.

Column	Min	Mean	Median	Max	Std. Dev.	Skewness	Kurtosis	No. Missing	No. +∞	No. -∞	Histogram
fixed acidity	4,6	8,4163	?	15,9	1,7421	0,9399	1,0835	0	0	0	
volatile acidity	0,12	0,5261	?	1,58	0,1798	0,7197	1,3463	0	0	0	
citric acid	0.0	0,2751	?	1	0,1955	0,2753	-0,8112	0	0	0	
residual sugar	0,9	2,5441	?	15,5	1,4052	4,4708	28,0633	0	0	0	

Data Preparation – Linear Correlation



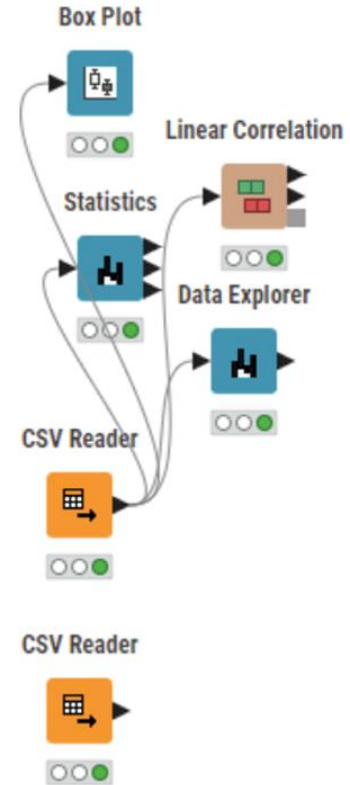
Calculates pairwise **linear correlation coefficients** between numerical columns in your dataset

■ corr = -1	fixed acidity	volatile a...	citric acid	residual s...	chlorides	free sulfu...	total sulf...	density	pH	sulphates	alcohol	quality
■ corr = +1												
✕ corr = n/a												
fixed acidity	■	■	■			■	■	■	■			✕
volatile acidity	■	■	■							■	■	✕
citric acid	■	■	■						■			✕
residual sugar				■								✕
chlorides					■				■	■	■	✕
free sulfur dioxide	■					■	■					✕
total sulfur dioxide	■					■	■				■	✕
density	■							■	■	■	■	✕
pH	■		■		■				■	■		✕
sulphates		■								■		✕
alcohol		■			■	■	■	■			■	✕
quality	✕	✕	✕	✕	✕	✕	✕	✕	✕	✕	✕	■

Data Preparation – Box Plot

In the Node Repository, search for the **Box Plot** node and drag into the workflow editor.

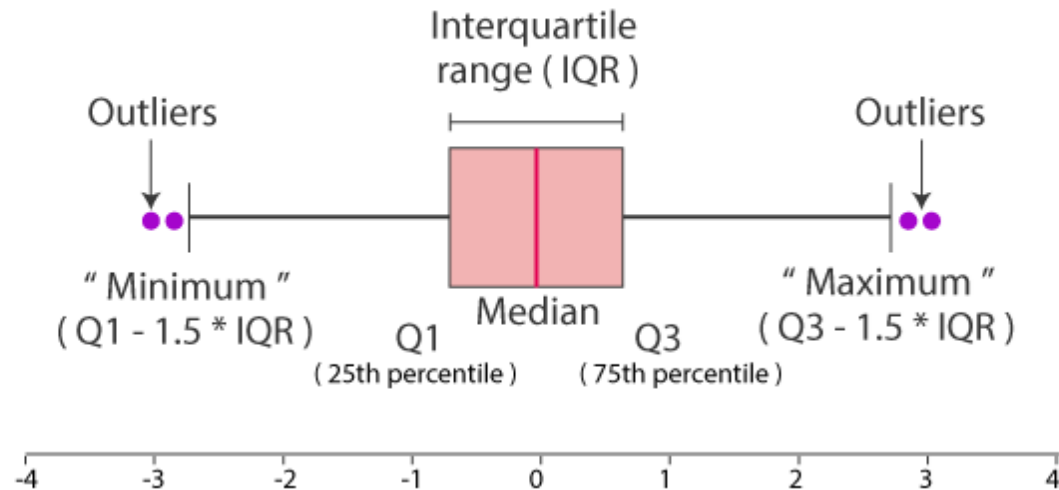
Connect the **output** port of the **CSV Reader** node, related to the **training** set, to the **input** port of the **Box Plot** node.



Data Preparation – Box Plot

- A graph summarising a set of data. The shape shows how the data is distributed and also any **outliers**.
- The line splitting the box in two represents the **median value**. 50% of the data lies on the left hand side and 50% lies on the right hand side.
- The left edge of the box represents the **lower quartile**: the value at which the first 25% of the data falls up to.
- The right edge of the box shows the **upper quartile**: 25% of the data lies to the right of the upper quartile value.
- The single points on the diagram show the **outliers**.

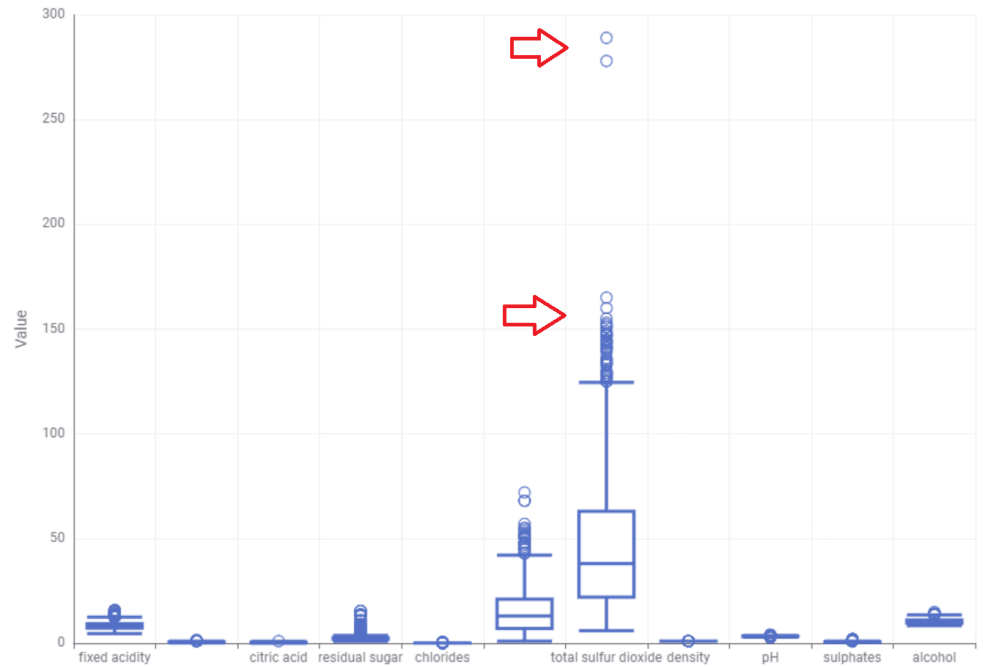
Data Preparation – Box Plot



Data Preparation – Box Plot

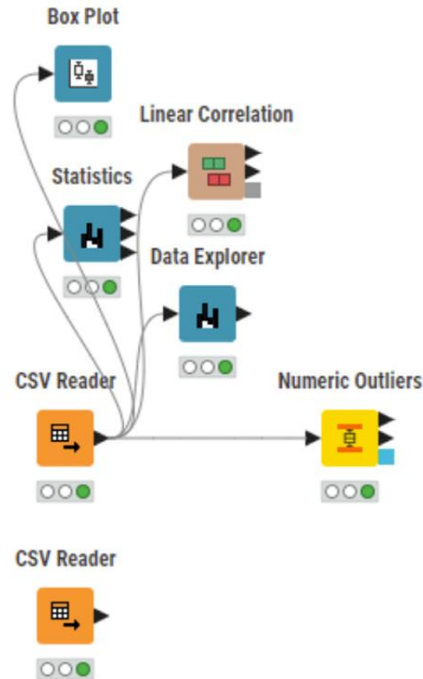


Displays key statistics of the data, such as the **median**, **quartiles**, and **potential outliers**.





Data Preparation – Numeric Outliers



In the Node Repository, search for the **Numeric Outliers** node and drag into the workflow editor.

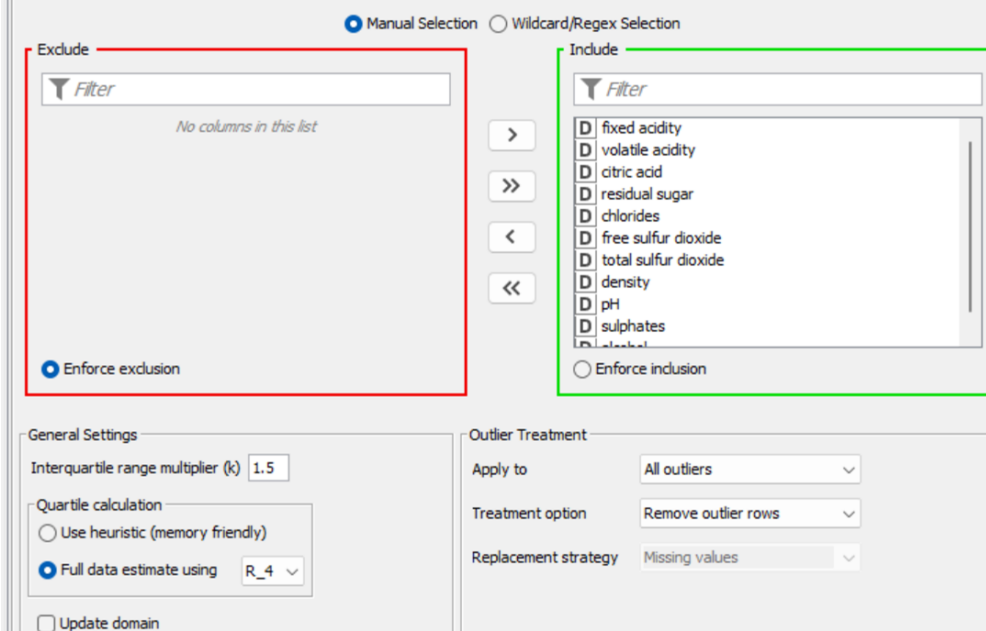
Connect the **output** port of the **CSV Reader node** to the **input** port of the **Numeric Outliers** node.

Data Preparation – Numeric Outliers



Open the node's configuration panel.

As Treatment Option, **choose Remove Outlier rows**. Click Ok and run the node.



The screenshot shows the 'Manual Selection' configuration panel for numeric outliers. The panel is divided into two main sections: 'Exclude' (highlighted with a red border) and 'Include' (highlighted with a green border). The 'Exclude' section has a 'Filter' input field and a list of columns. The 'Include' section has a 'Filter' input field and a list of columns. The 'General Settings' section includes 'Interquartile range multiplier (k)' set to 1.5, 'Quartile calculation' with 'Full data estimate using R_4' selected, and 'Update domain' unchecked. The 'Outlier Treatment' section includes 'Apply to' set to 'All outliers', 'Treatment option' set to 'Remove outlier rows', and 'Replacement strategy' set to 'Missing values'.

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

No columns in this list

☒ Enforce exclusion

Include

Filter

> >> < <<

☐ Enforce inclusion

General Settings

Interquartile range multiplier (k) 1.5

Quartile calculation

☐ Use heuristic (memory friendly)

☒ Full data estimate using R_4

☐ Update domain

Outlier Treatment

Apply to All outliers

Treatment option Remove outlier rows

Replacement strategy Missing values

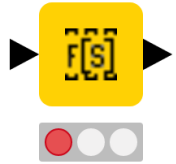
Data Preparation – String Manipulation

The quality attribute represents a **numerical** score, but the **values are prefixed with an equal sign (=)**, which can interfere with proper manipulation and analysis.

Since this attribute is the **target variable**, it's crucial to **clean the data** by **removing the equal sign**. This ensures that the values are treated as numerical data, allowing for correct model training and performance evaluation.

alcohol <i>Number (dou...</i>	quality <i>String</i>
9.4	=5
9.8	=5
9.8	=5
9.8	=6
9.4	=5
9.4	=5

Data Preparation – String Manipulation

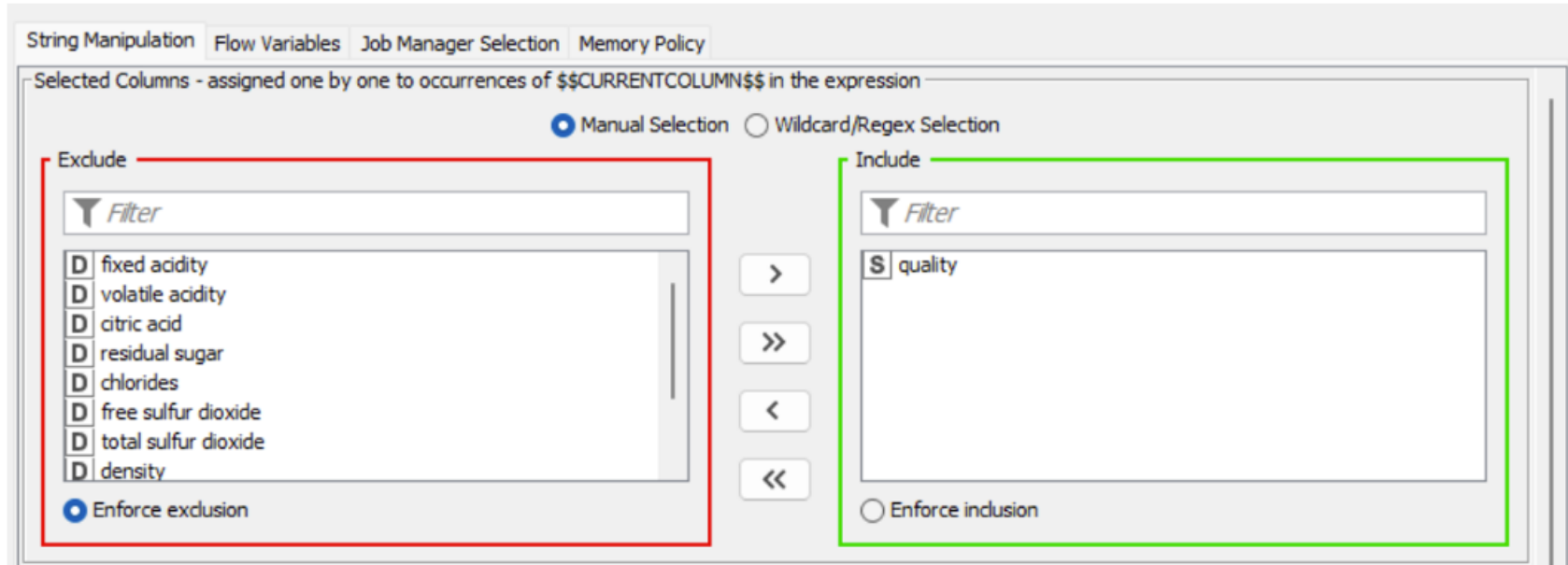


Search for the **String Manipulation (Multi Column)** node and drag into the workflow editor.

Connect the **output** port of the **Numeric Outliers** node to the **input** port of the **String Manipulation (Multi Column)** node.

Performs a wide range of operations on **string columns**, such as modifying or concatenating text data.

Data Preparation – String Manipulation



String Manipulation Flow Variables Job Manager Selection Memory Policy

Selected Columns - assigned one by one to occurrences of \$\$CURRENTCOLUMN\$\$ in the expression

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

- D fixed acidity
- D volatile acidity
- D citric acid
- D residual sugar
- D chlorides
- D free sulfur dioxide
- D total sulfur dioxide
- D density

☒ Enforce exclusion

Include

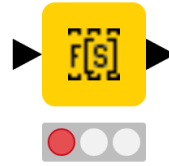
Filter

- S quality

☐ Enforce inclusion

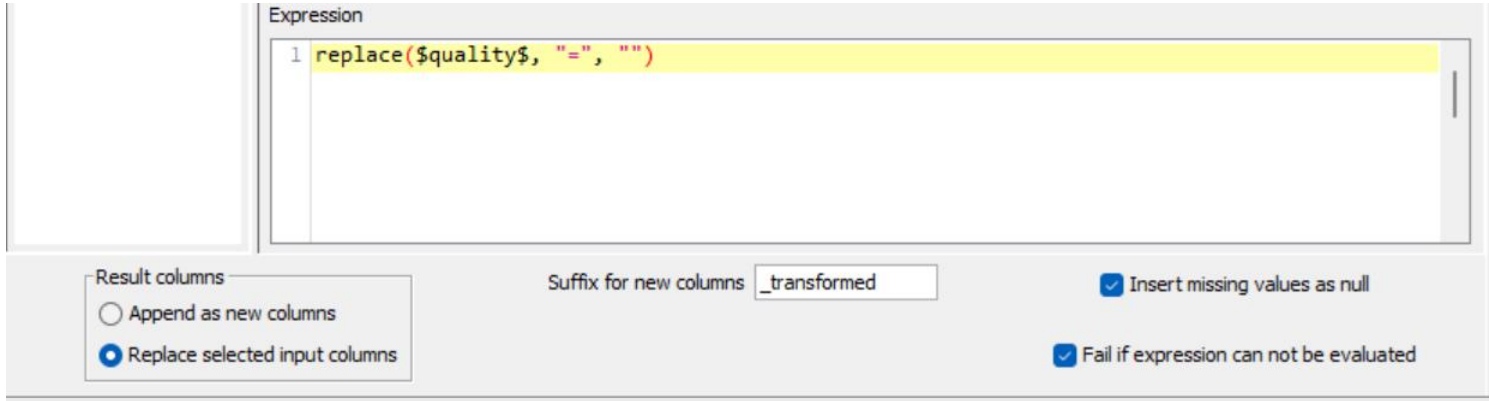
> >> < <<

Open the node's configuration panel.
Make sure to only include the **quality** attribute.



Performs a wide range of operations on **string columns**, such as modifying or concatenating text data.

Data Preparation – String Manipulation



The screenshot shows a software interface for data manipulation. At the top, there is a text area labeled "Expression" containing the code `1 replace($quality$, "=", "")`. Below this, there are several options for handling the result columns:

- Result columns:** Two radio buttons are present: "Append as new columns" (unselected) and "Replace selected input columns" (selected).
- Suffix for new columns:** A text input field containing the value `_transformed`.
- Checkboxes:** Two checkboxes are checked: "Insert missing values as null" and "Fail if expression can not be evaluated".

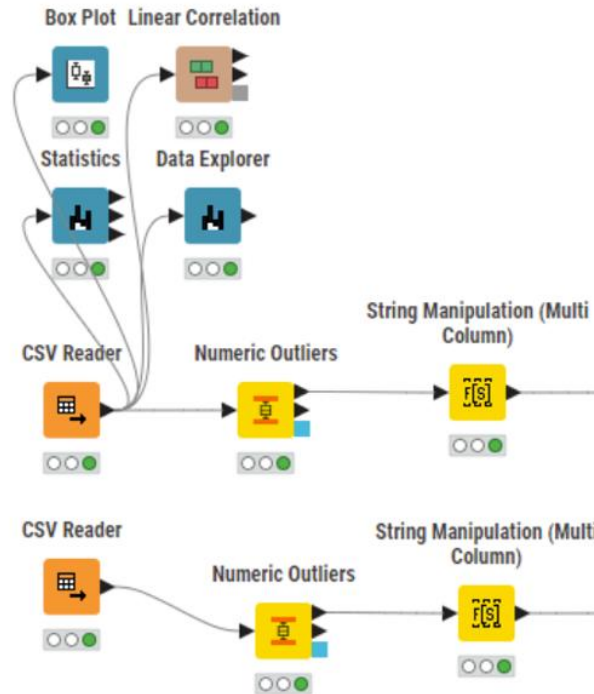
This expression will replace the “=” symbol by an **empty string**. Select the “**Replace selected input columns**” option to apply the changes to the column at hand.

Data Preparation – String Manipulation

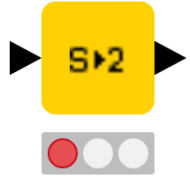
Click Ok and run the node. Observe the changes in the output panel.

alcohol <i>Number (dou...</i>	quality <i>String</i>
11.5	6
9.6	5
9.8	5
11	6
9.95	3

Pre-Processing Workflow



Data Preparation – String to Number



Converts
string columns
into numeric
data types

In a **classification** problem, the goal is to assign a binary output to each input. For example, to classify **whether** a wine is **good or not**, based on its features.

To do this, it would be necessary to **binarize** the **quality** column in the dataset, which contains numerical ratings.

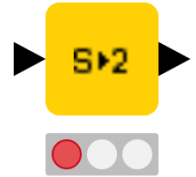
By binarizing the quality column, the problem can be simplified to be able to use classification algorithms that require a binary output.

Data Preparation – String to Number

However, quality ratings depend on various factors such as personal preferences, food pairings, or cultural backgrounds.

In this case, let's keep the **threshold at 5** and label **any wine with a quality rating above 5 as “high” (1)** quality and **any wine below 5 as “low” quality (0)**.

To do this transformation using the **Rule Engine** node in KNIME, the **quality** attribute must first be **converted** into a **numeric type**, ensuring that the rule logic can be applied correctly.



Converts
string columns
into numeric
data types

Data Preparation – String to Number

Column Selection

Column selection

Manual Wildcard Regex

Search Aa

Excludes

No columns in this list

Includes

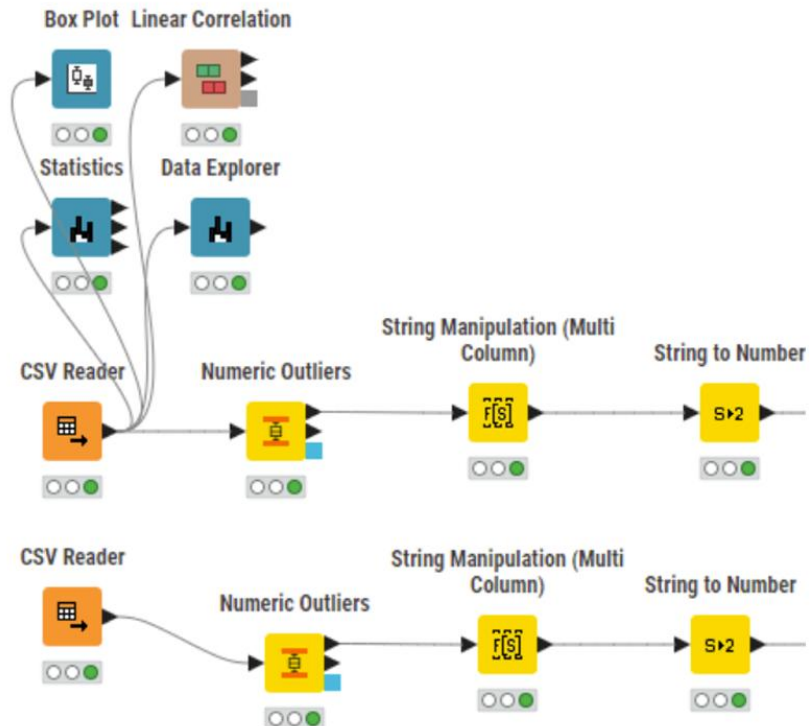
quality

Any unknown columns

alcohol Number (double)	quality Number (double)
9.4	5
9.8	5
9.8	5
9.8	6
9.4	5

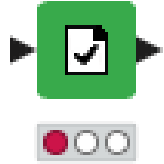
Make sure the quality attribute is included in the node's configuration panel. In the output panel, you can see that the **quality** attribute's type is now a **number**.

Data Preparation – Pre-Processing



Data Preparation - Rule Engine

Rule Engine



Creates new columns
or **modifies** existing
ones, based on if-
then-else rules

Connect the output port of the
String to Number node to the
input port of the Rule Engine
node.

Data Preparation - Rule Engine

Open the node's configuration panel and **write the expression** that will binarize the quality column.

Select the **Replace Column** option and choose the **quality attribute**.

Expression

?	1	// enter ordered set of rules, e.g.:
?	2	// \$double column name\$ > 5.0 => "large"
?	3	// \$string column name\$ LIKE "*blue*" => "small and blue"
?	4	// TRUE => "default outcome"
I	5	\$quality\$ > 5.0 => 1
I	6	\$quality\$ <= 5.0 => 0

☐ Append Column: prediction

☒ Replace Column: D quality

Data Preparation - Rule Engine

Run the node and check its output.

alcohol <i>Number (dou...</i>	quality <i>Number (inte...</i>
9.4	0
9.8	0
9.8	0
9.8	1
9.4	0

Data Preparation - Normalizer

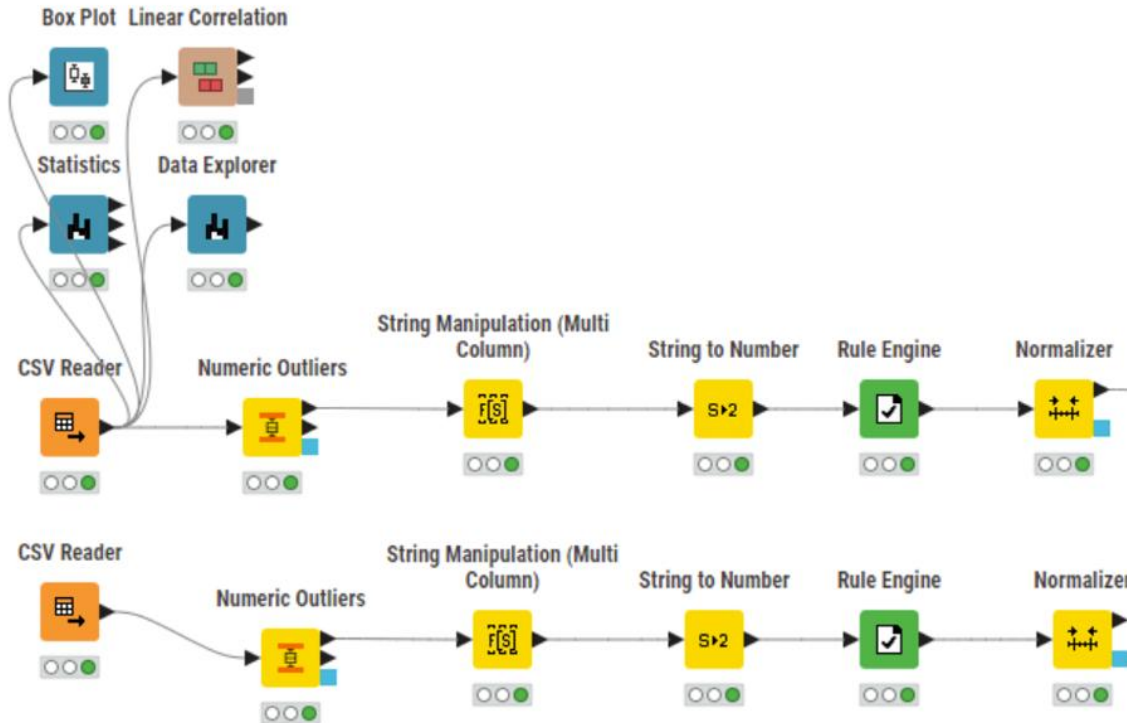


**Rescales or
standardizes**
numerical data to a
specific
range or distribution.

Since the data contains **different numerical parameters** (such as acidity levels, pH values, and alcohol content) for each feature, **normalization** transforms the input features to a **common scale** to ensure that they have **similar ranges**.

This can help to improve the performance of some machine learning algorithms, such as those that use distance-based metrics as in classification.

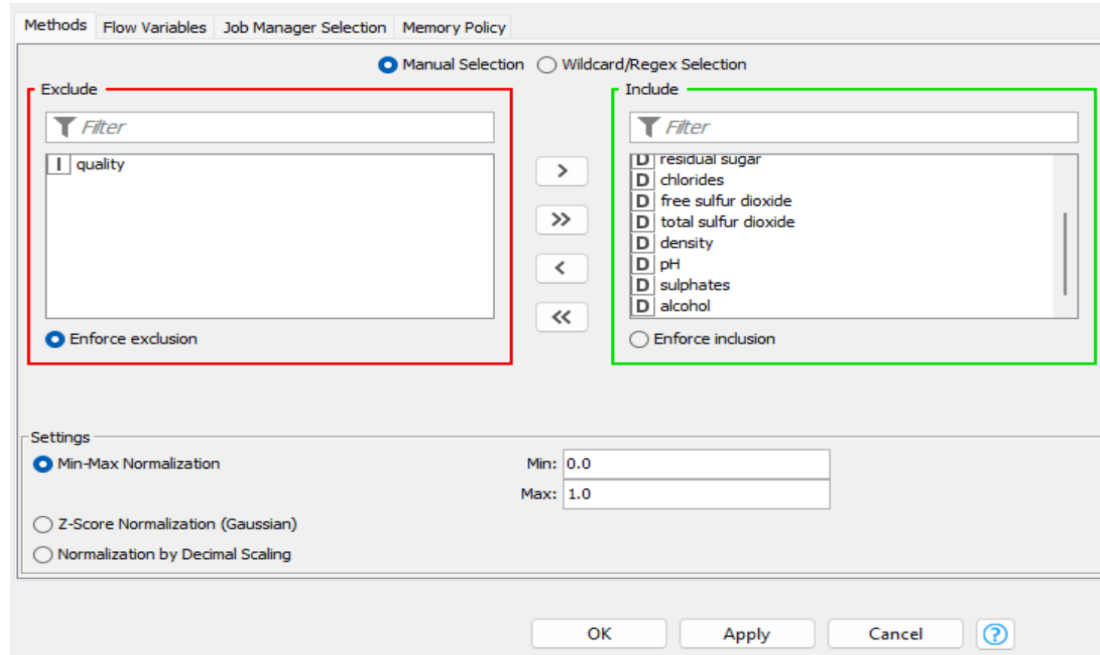
Data Preparation - Normalizer



Data Preparation - Normalizer

Exclude the **quality** attribute, since it is already a binary column.

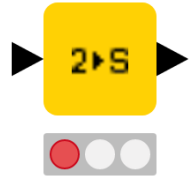
Apply the **Min-Max** normalization.



The screenshot shows the 'Data Preparation - Normalizer' dialog box with the following settings:

- Methods:** Flow Variables, Job Manager Selection, Memory Policy
- Selection:** ☒ Manual Selection, ☐ Wildcard/Regex Selection
- Exclude:** ☒ Enforce exclusion. The list contains 'quality'.
- Include:** ☐ Enforce inclusion. The list contains: residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol.
- Settings:** ☒ Min-Max Normalization, ☐ Z-Score Normalization (Gaussian), ☐ Normalization by Decimal Scaling. Min: 0.0, Max: 1.0.
- Buttons:** OK, Apply, Cancel, ?

Data Preparation – Number to String



Converts
numeric
columns into
string format.

The Number to String node is a perfectly valid and effective way to convert the binary target values (0 and 1) into string format for the Support Vector Machines Learner.

Data Preparation – Number to String

Columns  

Manual Wildcard Regex

Aa

Excludes

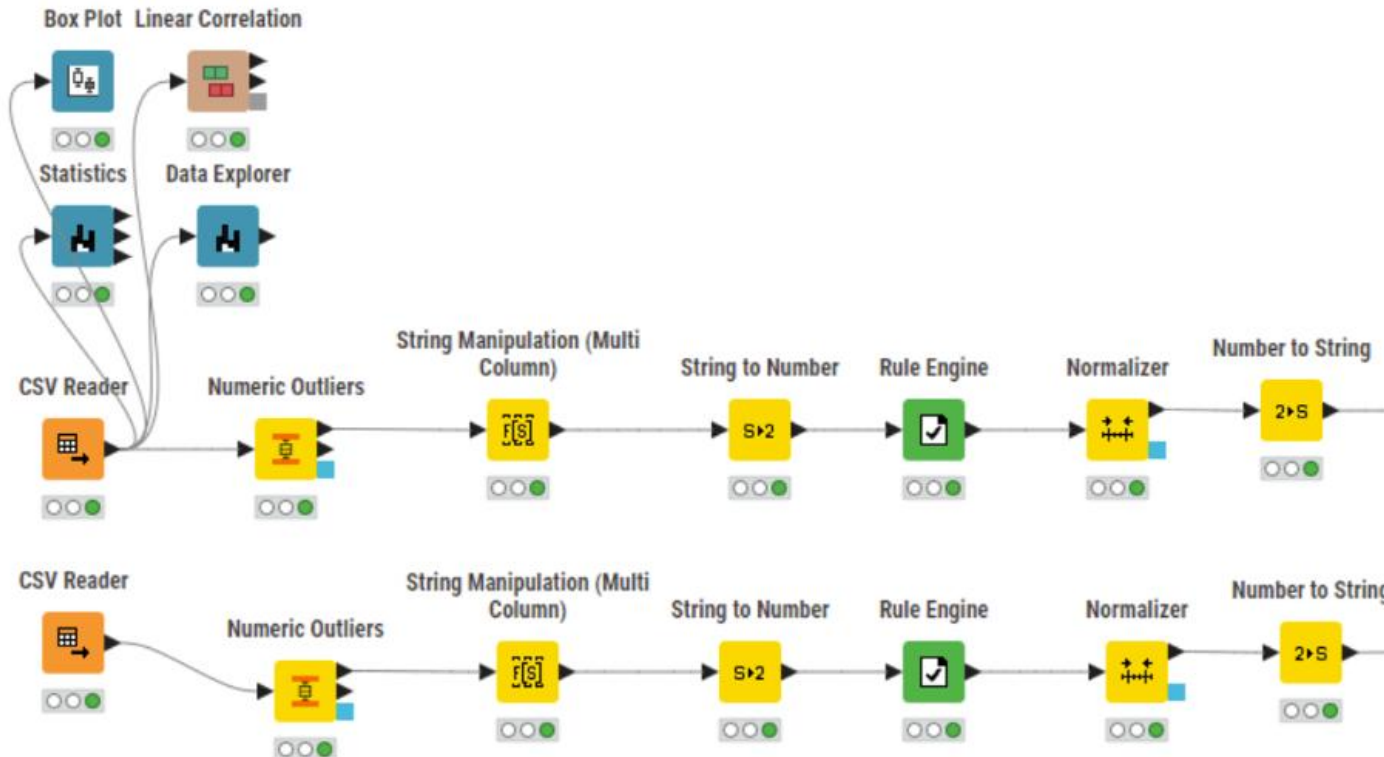
fixed acidity
volatile acidity
citric acid
residual sugar
chlorides
free sulfur dioxide
Any unknown columns

Includes

quality

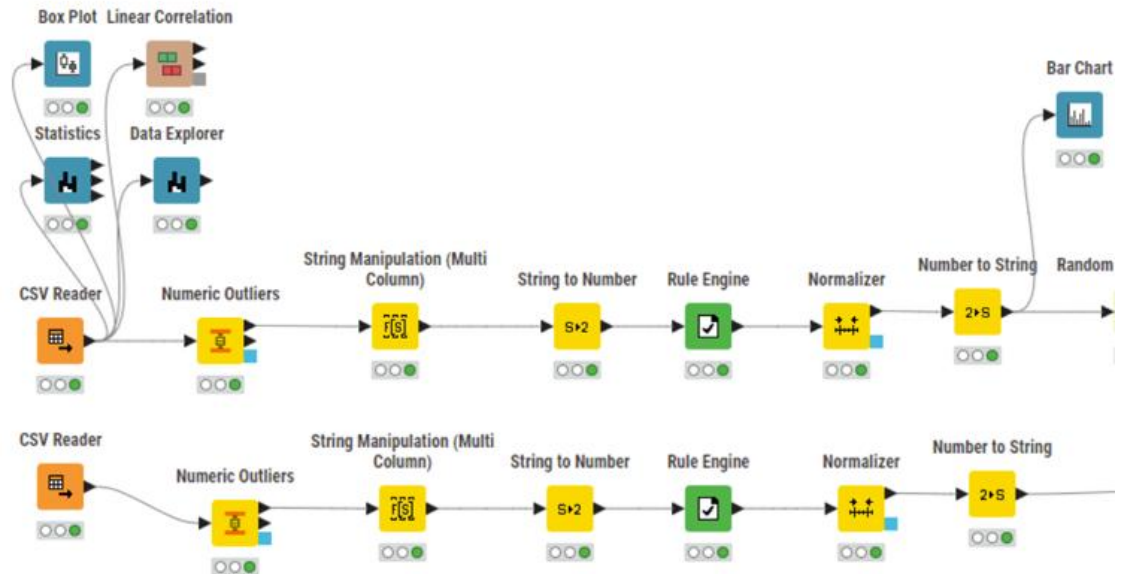
alcohol <i>Number (dou...</i>	quality <i>String</i>
0.17	1
0.106	0
0.255	0
0.383	1

Data Preparation – Number to String



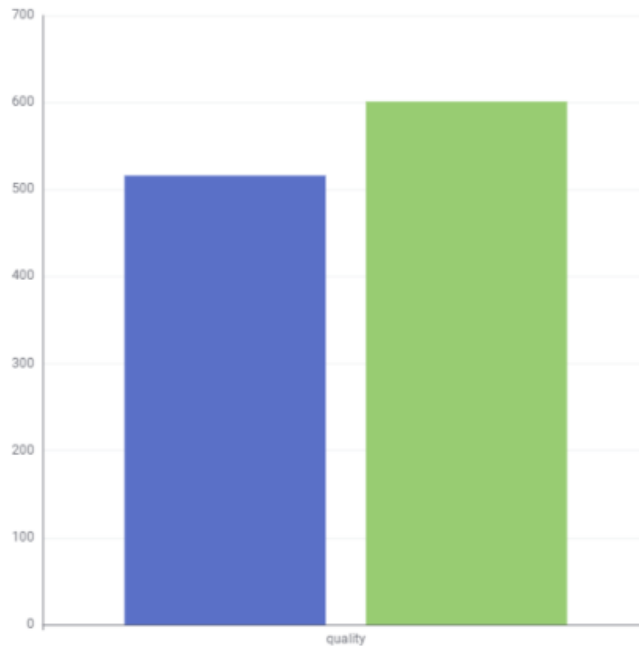
Data Preparation – Bar Chart

Add a Bar Chart node after the Number to String node, related to the **training set**.



Data Preparation

While the difference in class sizes isn't extreme, there is still a notable disparity between the two classes as the high quality class has more samples than the low quality one.



Category dimension 

quality 

Aggregation 

☐ None ☒ Occurrence count ☐ Sum

☐ Average

Frequency dimensions 

Manual Wildcard Regex Type

Search

Excludes

- fixed acidity
- volatile acid...
- citric acid
- residual sug...
- chlorides
- free sulfur d...
- Any unknown columns

Includes

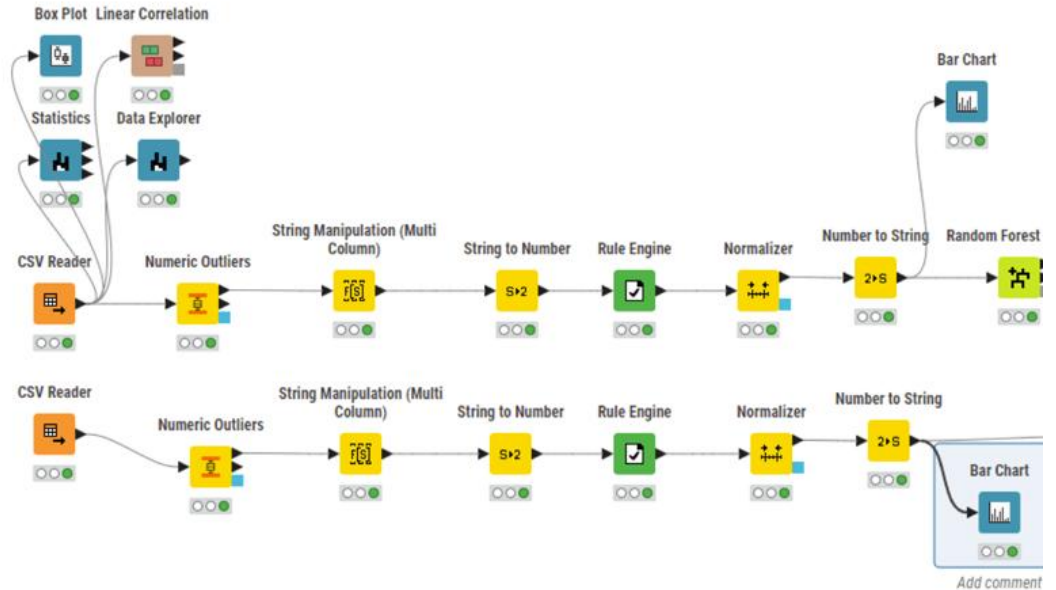
No columns in this list

Plot

Title

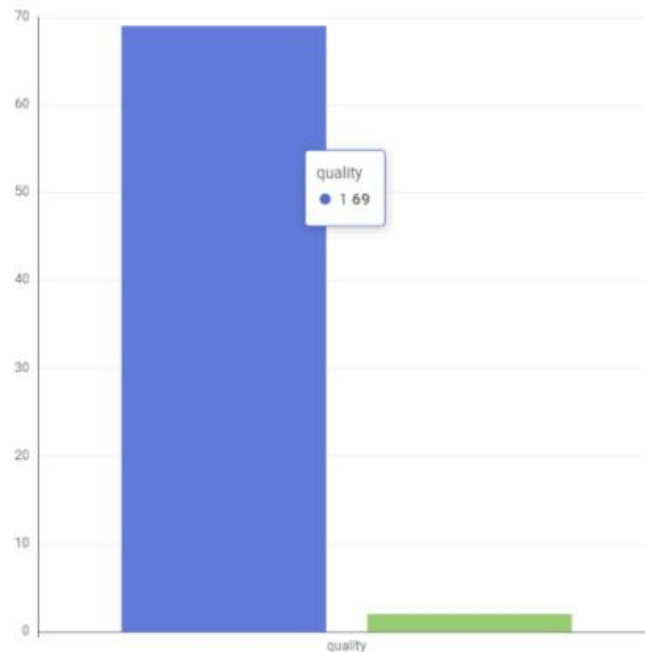
Data Preparation

Add a Bar Chart node after the Number to String node, related to the **testing set**.



Data Preparation

The test set has a majority of high-quality samples, which can significantly affect the evaluation of the model's performance.



Category dimension 

quality 

Aggregation 

☐ None ☒ Occurrence count ☐ Sum

☐ Average

Frequency dimensions 

Manual Autocard Pages Type

Search

Excludes

- fixed acidity
- volatile acid...
- citric acid
- residual sug...
- chlorides
- free sulfur d...
- why unknown parameter

Includes

No column in this set

Plot

Title



KNIME Analytics Platform

Goal:

- Understand **Support Vector Machine**
- Develop a **SVM model** capable of **classifying wine's quality** based on several properties such as pH, density, acidity, etc
- Analyze the **performance** of the final model

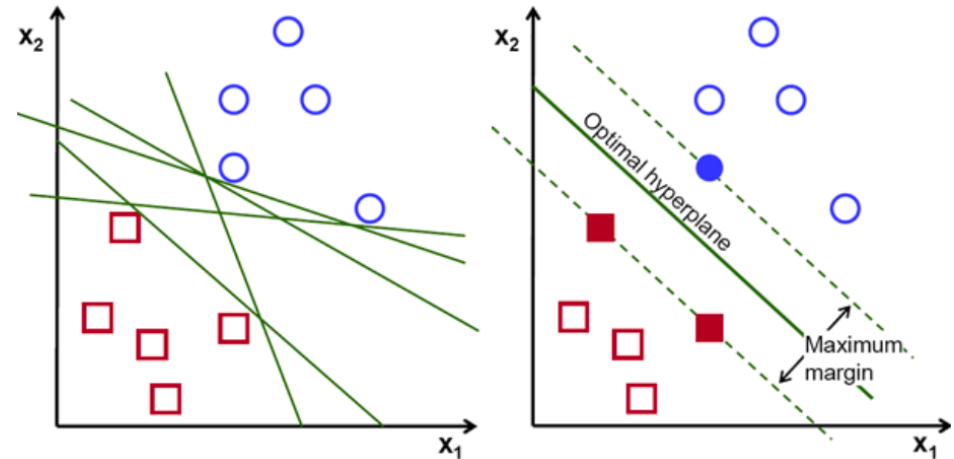
SVM

Support Vector Machine is a supervised Machine Learning algorithm that can be used for both **classification** (mostly) or regression problems.

The main idea is to plot each data item as a point in an n -dimensional space (n is the number of features), performing classification by **finding the hyper-plane that differentiates the classes.**

SVM

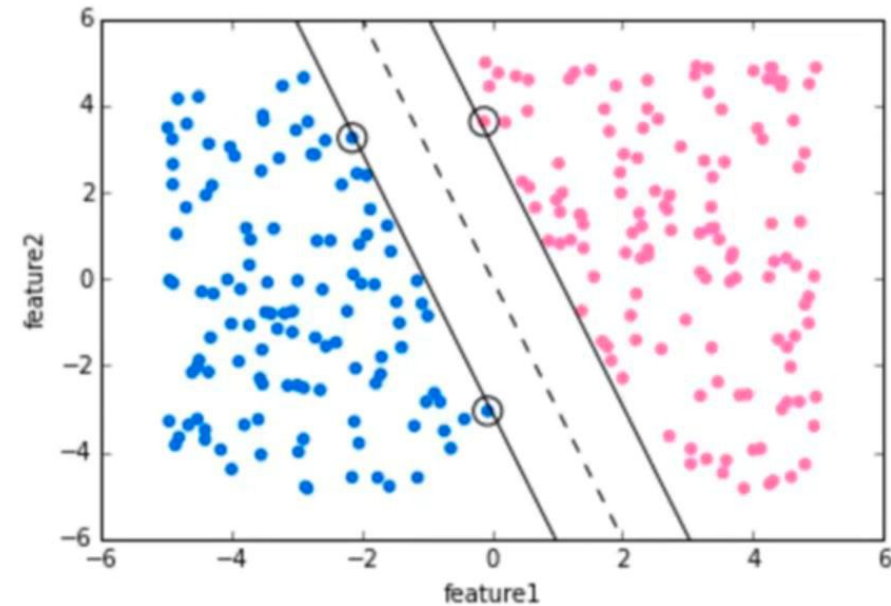
- Works well for **classifying higher-dimensional data** (datasets with lots of features)
- Finds higher-dimensional support vectors which “divides” the data
- **Applies kernels to represent data in higher-dimensional spaces** to find hyperplanes that might not be apparent in lower dimensions.



SVM

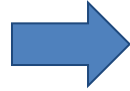
How it works?

1. Image a labelled training data
2. Draw a separating “hyperplane” between the classes –many options that separate perfectly...
3. Choose a hyperplane that maximizes the margin between classes –vector points that the margin lines touch are known as Support Vectors



Exercise:

➤ **Problem:** Development of a SVM model capable of **classifying wine's quality as high or low**.



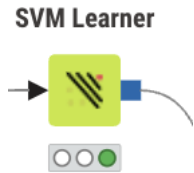
Dataset: Table with wine's properties:

- **fixed acidity:** Total amount of organic acids found in wine.
- **volatile acidity:** Quantity of acetic acid generated by yeast or bacteria.
- **citric acid:** The quantity of citric acid.
- **residual sugar:** The sugar remaining in wine after fermentation.
- **chlorides:** Amount of salt (chlorides) in the wine.
- **free sulfur dioxide:** Amount of free sulfur dioxide in the wine.
- ...

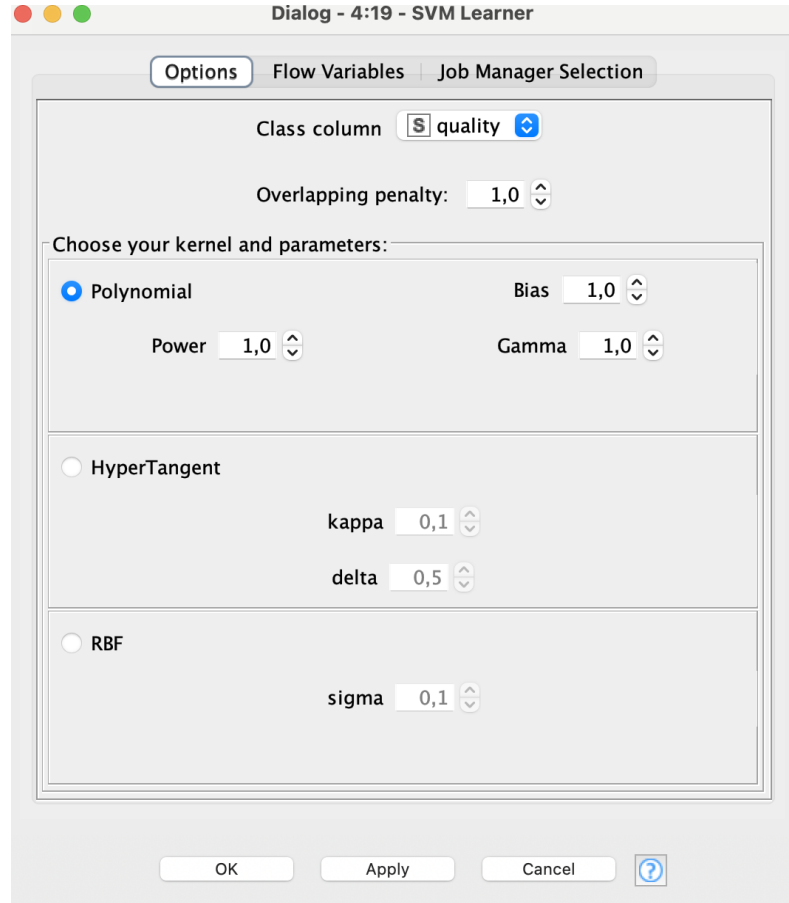
Exercise:

First Step: Train the model

Node: SVM Learner



Trains a SVM model

A screenshot of the 'Dialog - 4:19 - SVM Learner' window. It has three tabs: 'Options', 'Flow Variables', and 'Job Manager Selection'. The 'Options' tab is active. It shows 'Class column' set to 'quality' with a dropdown arrow. Below it, 'Overlapping penalty' is set to '1,0' with a spinner. A section titled 'Choose your kernel and parameters:' contains three options: 'Polynomial' (selected), 'HyperTangent', and 'RBF'. Under 'Polynomial', 'Power' is '1,0' and 'Gamma' is '1,0'. Under 'HyperTangent', 'kappa' is '0,1' and 'delta' is '0,5'. Under 'RBF', 'sigma' is '0,1'. At the bottom are 'OK', 'Apply', 'Cancel', and a help icon.

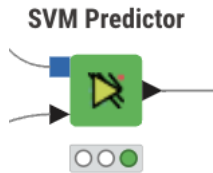
Choose the target variable:

quality

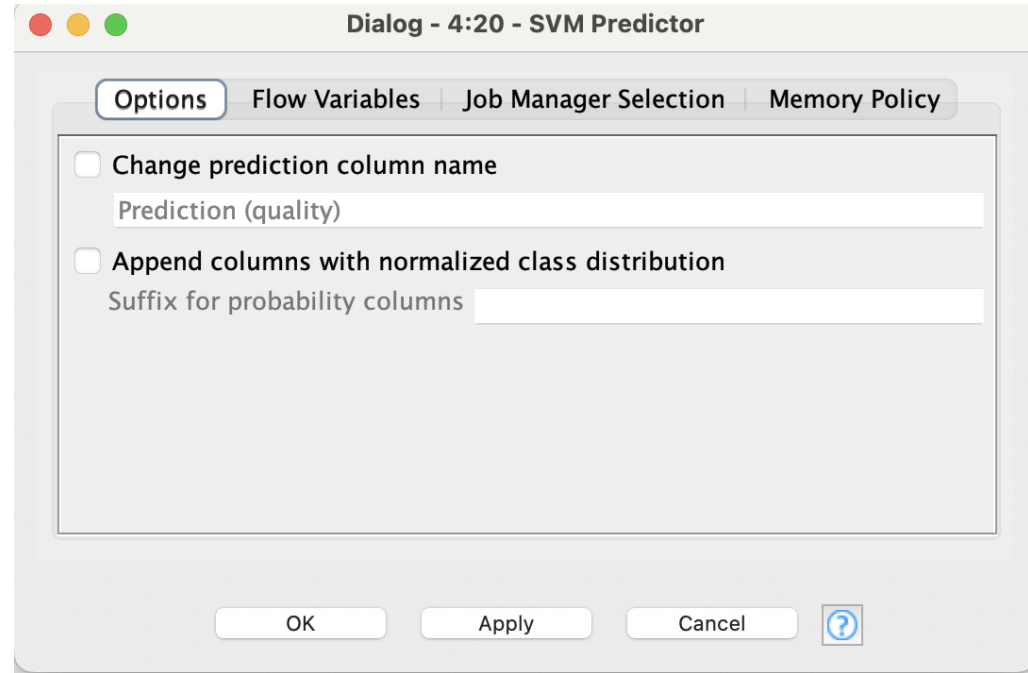
Exercise:

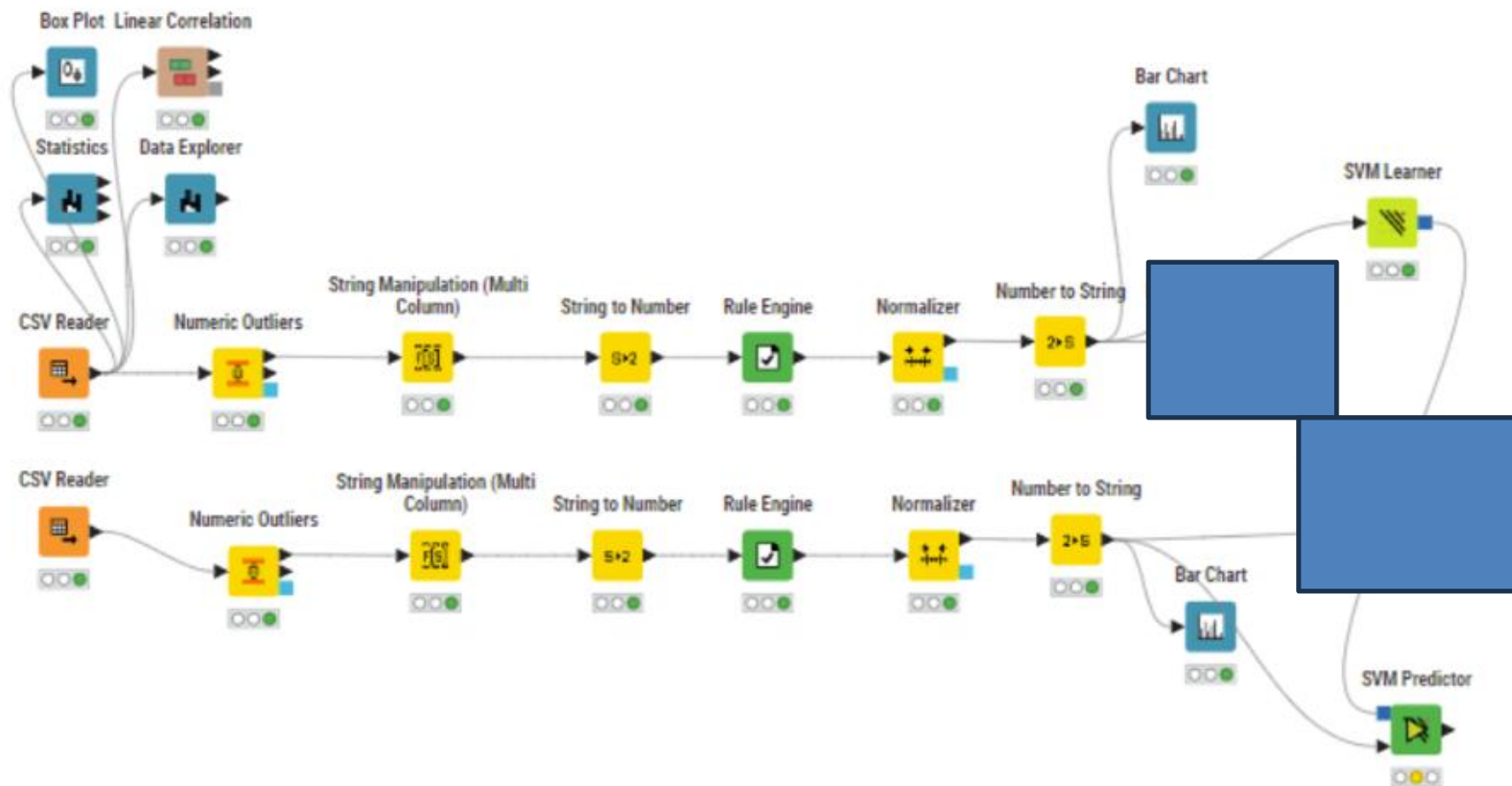
Second Step: Test the model

Node: SVM Predictor



*Tests a previously trained
SVM model*





Exercise:

Final Step: Evaluate the model

Measure:

Accuracy



- Calculates the **ratio** of **accurate predictions** to **total predictions** made by the model.
- A **high accuracy** score means the model is making **mostly correct predictions**.
- A **low accuracy** score means the model is making **mostly incorrect predictions**.

Exercise:

Final Step: Evaluate the model

Measure:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$



- **TP:** The number of positive instances that are correctly classified as positive by the model.
True Positives.
- **FP:** The number of negative instances that are incorrectly classified as positive by the model.
False Positives.
- **TN:** The number of negative instances that are correctly classified as negative by the model.
True Negatives.
- **FN:** The number of positive instances that are incorrectly classified as negative by the model.
False Negatives.

Exercise:

Final Step: Evaluate the model

Measure:

Confusion matrix



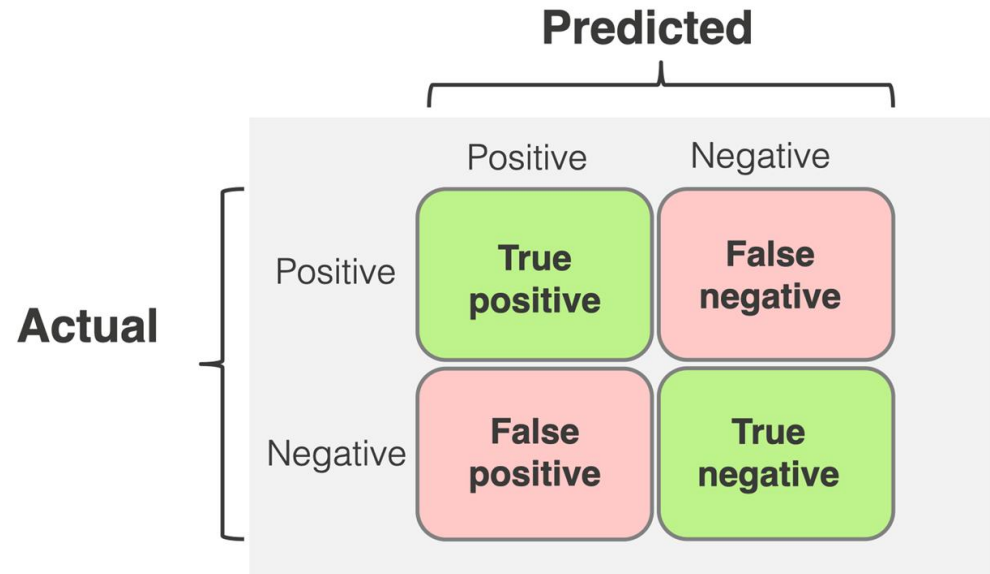
- **Chart** that presents the outcomes of a **binary classifier** in a summary format.
- It gives a **thorough examination** of the model's classification performance by displaying the counts of TP, TN, FP, and FN.

Exercise:

Final Step: Evaluate the model

Measure:

Confusion matrix



Exercise:

Final Step: Evaluate the model

Measure:

Cohen's Kappa



- Assesses the **level of agreement** between two classification models, accounting for the agreement that could happen by chance.
- Evaluates how well the model's predictions match the actual labels, especially when the dataset may be imbalanced.
- **1: Perfect agreement** between the predicted and actual labels.
- **0: Agreement is no better** than random chance.
- **Negative values: Worse-than-random agreement** (the model consistently disagrees with the actual labels).

Exercise:

Final Step: Evaluate the model

Node: Scorer



Provides various metrics such as accuracy and confusion matrix to assess the model's effectiveness.

File

Scorer Flow Variables Job Manager Selection Memory Policy

First Column
[S] quality

Second Column
[S] Prediction (quality)

Sorting of values in tables
Sorting strategy: Insertion order ☐ Reverse order

Provide scores as flow variables
☐ Use name prefix

Missing values
In case of missing values: ☒ Ignore ☐ Fail

OK Apply Cancel ?

Exercise:

Final Step: Evaluate the model

Node: Scorer



*Provides various metrics
such as accuracy and
confusion matrix to assess
the model's effectiveness.*

File Hilite

quality \ Pr...	1	0
1	49	20
0	0	2

Correct classified: 51

Wrong classified: 20

Accuracy: 71,831%

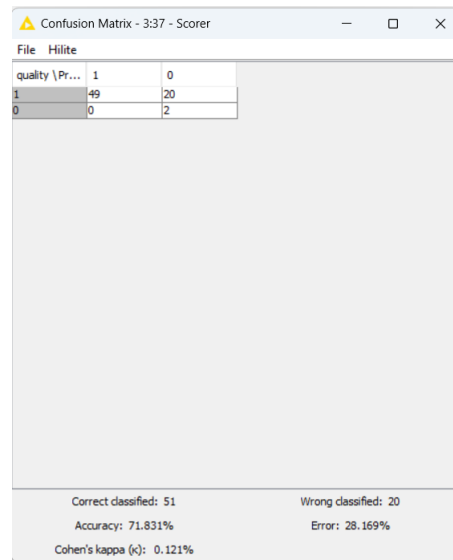
Error: 28,169%

Cohen's kappa (κ): 0,121%

Exercise:

Final Step: Evaluate the model

- The model **correctly classified 49 samples** as high quality (1), demonstrating its effectiveness in identifying high-quality instances.
- The model **incorrectly predicted 20 samples** as low quality (0) when they were actually high quality, indicating a tendency to misclassify some high-quality samples.
- The model incorrectly classified 2 low-quality samples as high (1), and none correctly as low (0).



Confusion Matrix - 3:37 - Scorer

File Hilite

quality \ Pr...	1	0
1	49	20
0	0	2

Correct classified: 51
Accuracy: 71.831%
Cohen's kappa (κ): 0.121%

Wrong classified: 20
Error: 28.169%

Exercise:

Correct classified: 51

Wrong classified: 20

Accuracy: 71,831%

Error: 28,169%

Cohen's kappa (κ): 0,121%

Final Step: Evaluate the model

- An accuracy of **71.831%** indicates that it correctly classified approximately 71.8% of the samples in the dataset.
- A kappa of **0.121%** indicates that the model's predictions are not very significantly better than random guessing.

Correct classified: 51	Wrong classified: 20
Accuracy: 71,831%	Error: 28,169%
Cohen's kappa (κ): 0,121%	

Exercise:

Final Step: Evaluate the model

Insights:

- The accuracy appears reasonably high at first glance; however, the low kappa value suggests that the model **may be primarily predicting the majority class**.
- This scenario is common in **imbalanced datasets**, where one class significantly outnumbers the other.
- The test set has a **majority of high-quality samples**, which significantly affected the evaluation of the model's performance.

Exercise: Decision Tree Learner

Apply the Decision Tree Learner and Predictor Nodes to the pre-processed wine quality dataset and evaluate the model.

Exercise: Decision Tree Learner

Apply the Decision Tree Learner and Predictor Nodes to the pre-processed wine quality dataset and evaluate the model.

SOLUTION:

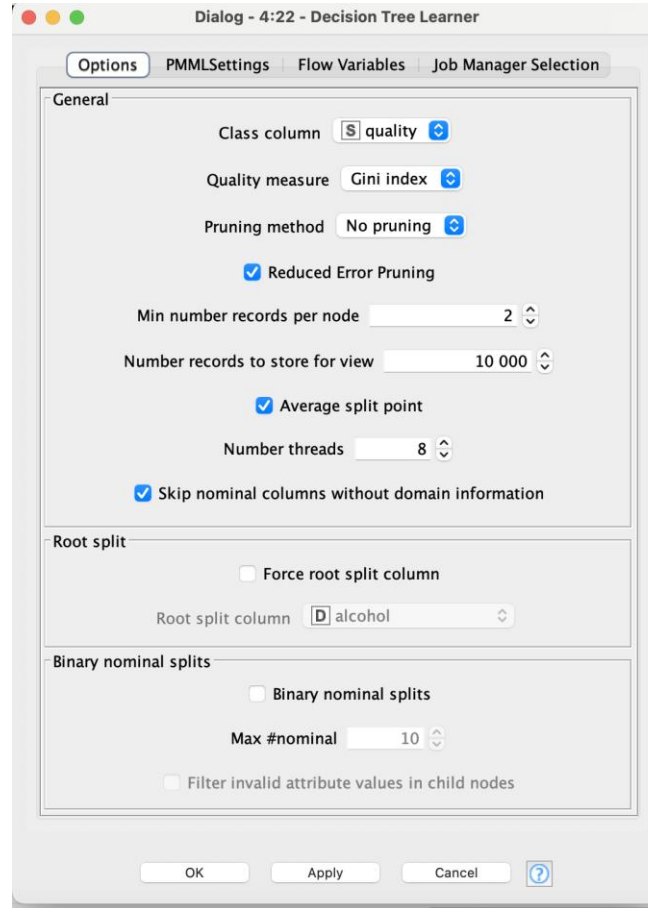
Exercise:

First Step: Train the model

Node: Decision tree Learner



Trains a Decision Tree model



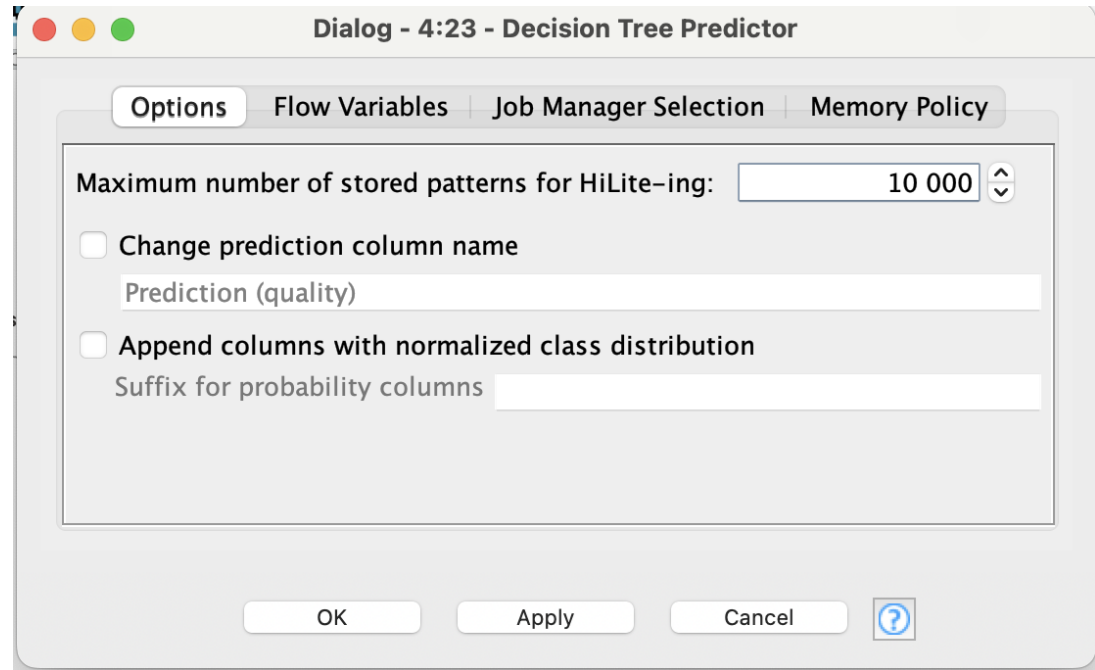
Exercise:

Second Step: Test the model

Node: Decision Tree Predictor



*Tests a previously trained
Decision Tree model*



Exercise:

Final Step: Evaluate the model

Measure:

Accuracy



- Calculates the **ratio** of **accurate predictions** to **total predictions** made by the model.
- A **high accuracy** score means the model is making **mostly correct predictions**.
- A **low accuracy** score means the model is making **mostly incorrect predictions**.

Exercise:

Final Step: Evaluate the model

Measure:

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- **TP:** The number of positive instances that are correctly classified as positive by the model.
True Positives.
- **FP:** The number of negative instances that are incorrectly classified as positive by the model.
False Positives.
- **TN:** The number of negative instances that are correctly classified as negative by the model.
True Negatives.
- **FN:** The number of positive instances that are incorrectly classified as negative by the model.
False Negatives.

Exercise:

Final Step: Evaluate the model

Measure:

Confusion matrix



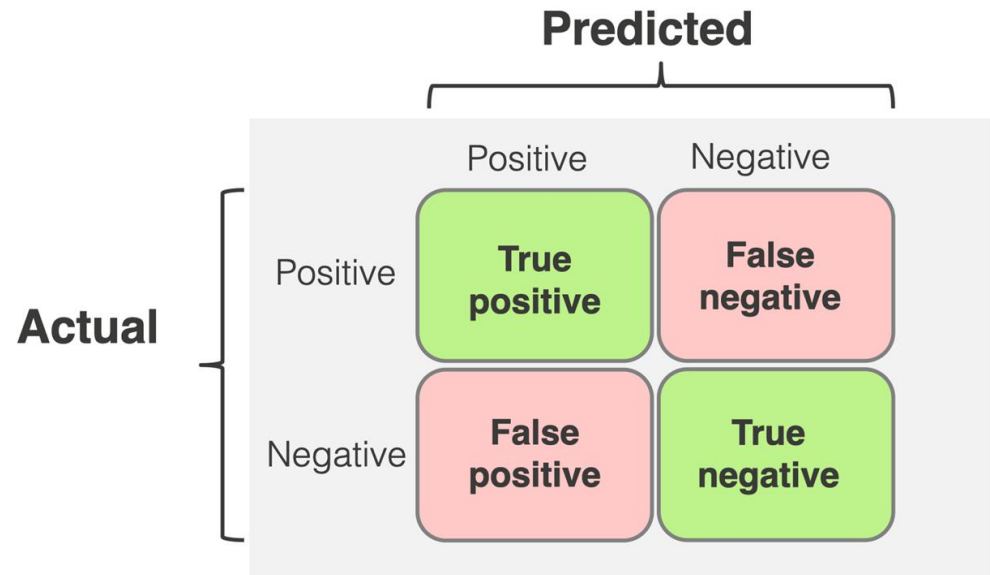
- **Chart** that presents the outcomes of a **binary classifier** in a summary format.
- It gives a **thorough examination** of the model's classification performance by displaying the counts of TP, TN, FP, and FN.

Exercise:

Final Step: Evaluate the model

Measure:

Confusion matrix



Exercise:

Final Step: Evaluate the model

Measure:

Cohen's Kappa



- Assesses the **level of agreement** between two classification models, accounting for the agreement that could happen by chance.
- Evaluates how well the model's predictions match the actual labels, especially when the dataset may be imbalanced.
- **1: Perfect agreement** between the predicted and actual labels.
- **0: Agreement is no better** than random chance.
- **Negative values: Worse-than-random agreement** (the model consistently disagrees with the actual labels).

Exercise:

Final Step: Evaluate the model

Node: Scorer



Provides various metrics such as accuracy and confusion matrix to assess the model's effectiveness.

File

Scorer Flow Variables Job Manager Selection Memory Policy

First Column
[S] quality

Second Column
[S] Prediction (quality)

Sorting of values in tables
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In case of missing values: ☒ Ignore ☐ Fail

OK Apply Cancel ?

Exercise:

Final Step: Evaluate the model

Node: Scorer



Provides various metrics such as accuracy and confusion matrix to assess the model's effectiveness.

Confusion Matrix - 4:24 - Scorer			
File	Hilite		
quality \ Pr...	1	0	
1	32	11	
0	17	11	
Correct classified: 43 Wrong classified: 28 Accuracy: 60,563% Error: 39,437% Cohen's kappa (κ): 0,142%			

Exercise:

Final Step: Evaluate the model

Confusion Matrix - 4:24 - Scorer

File

Hilite

quality \ Pr...	1	0	
1	32	11	
0	17	11	

Correct classified: 43

Wrong classified: 28

Accuracy: 60,563%

Error: 39,437%

Cohen's kappa (κ): 0,142%

- The model **correctly classified 32 samples** as high quality (1), demonstrating its effectiveness in identifying high-quality instances.
- The model **incorrectly predicted 11 samples** as low quality (0) when they were actually high quality, indicating a tendency to misclassify some high-quality samples.
- The model accurately classified 11 samples as low quality (0).

Exercise:

Final Step: Evaluate the model

Confusion Matrix - 4:24 - Scorer

File

Hilite

quality \ Pr...	1	0	
1	32	11	
0	17	11	

Correct classified: 43

Wrong classified: 28

Accuracy: 60,563%

Error: 39,437%

Cohen's kappa (κ): 0,142%

- An accuracy of **60.563%** indicates that it correctly classified approximately 60.6% of the samples in the dataset.
- A kappa of **0.142%** indicates that the model's predictions are not significantly better than random guessing.

Exercise:

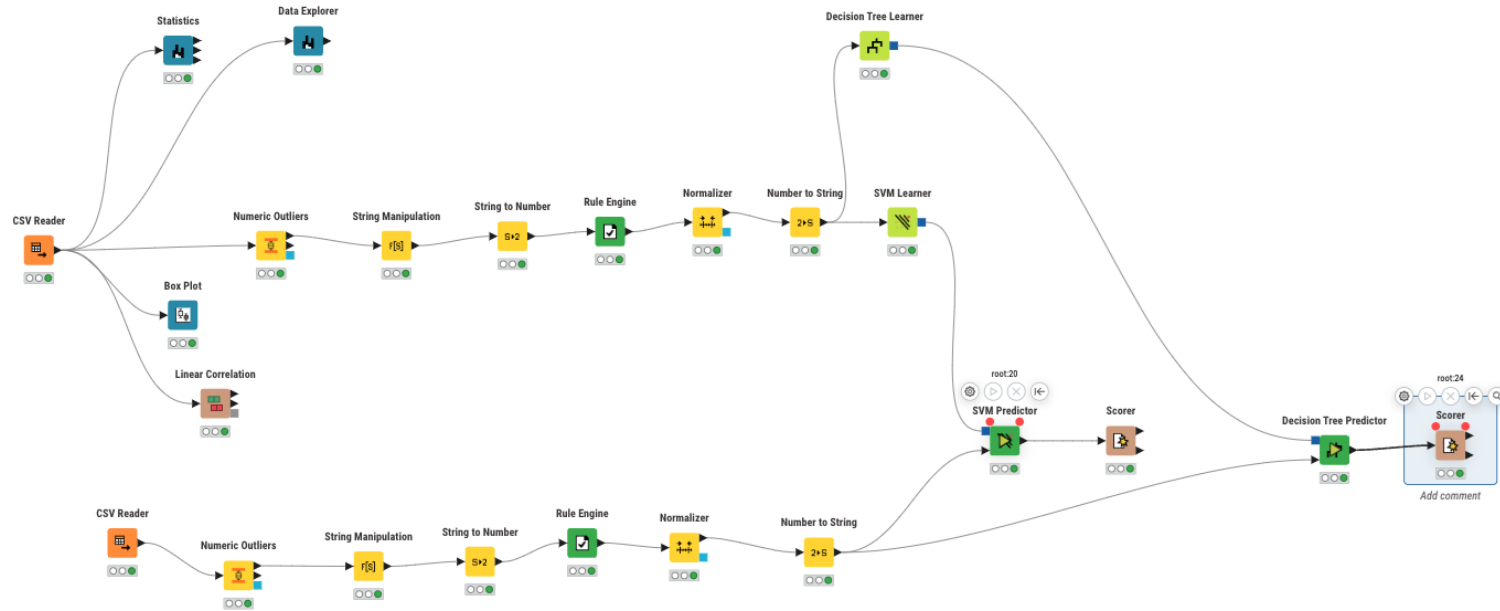
Final Step: Evaluate the model

Insights:

- The accuracy appears reasonably high at first glance; however, the low kappa value suggests that the model **may be primarily predicting the majority class**.
- This scenario is common in **imbalanced datasets**, where one class significantly outnumbers the other.
- The test set has a **majority of high-quality samples**, which significantly affected the evaluation of the model's performance.

Confusion Matrix - 4:24 - Scorer		
File	Hilite	
quality \ Pr...	1	0
1	32	11
0	17	11
Correct classified: 43		
Wrong classified: 28		
Accuracy: 60,563%		
Error: 39,437%		
Cohen's kappa (κ): 0,142%		

Final Workflow





Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Pre-Processing + SVM + Decision Tree

**MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA**

**Inteligência Artificial para as Telecomunicações
2025/2026**